

CSE 420

Assignment 3

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CSE420

ASSIGNMENT-3

Ans. to the Q.No.1

Here, $b = \text{func}(c, d + k * y)$

for this, the three address code,

$$t_1 = k * y$$

$$t_2 = d + t_1$$

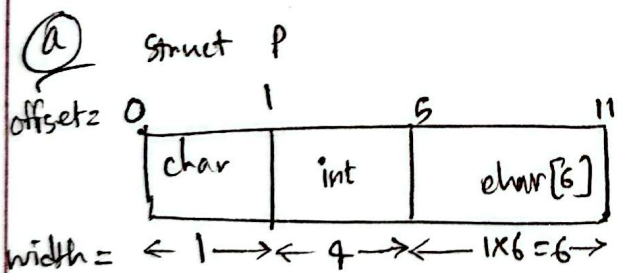
param c

param ~~c~~ t_2

$$t_3 = \text{call func}, 2$$

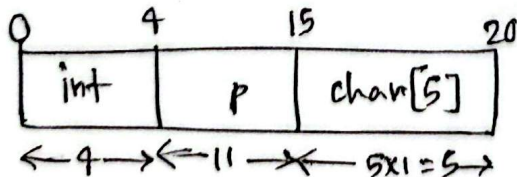
$$b = t_3$$

Ans. to the Q.No.2



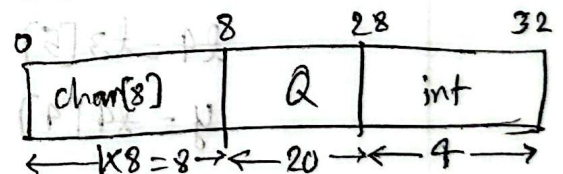
$$\text{So, total width for } = (1 + 4 + 6) = 11$$

Struct Q



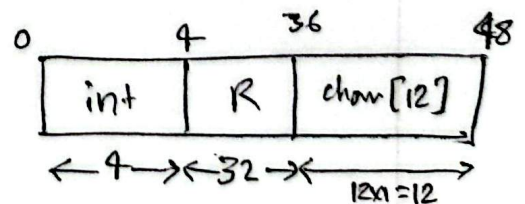
$$\therefore \text{So, total width} = (4 + 11 + 5) = 20$$

Struct R



$$\text{So total width} = 8 + 20 + 4 = 32$$

Struct S



$$\text{So total width} = 4 + 32 + 12 = 48$$

So, $y = \text{StudentRecord}.\text{info}.\text{course}.\text{result}.\text{section} [4]$

So, total offset = $(4+8+4+5+4) = 25$

info = 4
course = 8
result = 4
section = 5
section[4x1] = 4

int = 4 bytes
char = 1 bytes

②b) the equivalent three address code:

according to ①a, the offsets are,

info = 4, course = 8, result = 4, section = 5, section[4x1] = 4
(t1) (t2) (t3) (t4) (t5)

So,

$t1 = \text{@studentRecord}[4]$

$t2 = t1[8]$

$t3 = t2[4]$

$t4 = t3[5]$

$y = t4[4]$

(Ans)